



EN 507: Transport Phenomena for Energy Systems

Course Code	EN 507
Course Name	Transport Phenomena for Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/PG
Mutual Exclusion	None
Approval	9th Senate (May 2016)

Course Contents

Module I: Introduction to transport phenomena, Vectors and tensors, types/uses of control volume, notion of conservation principles and constitutive relations (4 Lectures)

Module II: Mechanisms of momentum transport, shell-momentum balance, equations of change for isothermal systems, applications of shell-momentum balance and equations of change to solve 1D problems in laminar flow, time dependent flow of Newtonian fluids, stream function and velocity potential, time smoothed equations for turbulent flows, boundary layer flow past different bodies (16 Lectures)

Module III: Thermal conductivity and energy transport, Fourier's law of heat conduction, shell-energy balance and boundary conditions, equations of change for non-isothermal systems, temperature distributions and unsteady heat conduction in solids, steady and unsteady free convection, temperature distributions in turbulent flow and turbulent heat flux, forced convection in tubes, submerged bodies and packed beds (14 Lectures)

Module IV: Mechanisms of mass transport, Fick's law of diffusion, shell-mass balance, equations of change for multi-component systems, laminar flow concentration profiles, steady and unsteady convective mass transport, diffusion in gases and liquids, multi-component mass diffusion, reaction rates, mechanisms, time scales (8 Lectures)

References

1. R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot, **Transport phenomena**, 2nd edition, Wiley, 2001.
2. J.C. Slattery, **Advanced transport phenomena**, Cambridge University Press, New York, 1999.
3. Welty J.R, Wicks J.E, Wilson R.E, Rorrer G, **Fundamentals of momentum, heat and mass transport**, 4th Edition, Wiley, 2001.